

Cross-part rainbow near-perfect matchings in $\mathbb{F}_3^2$: an exact finite proof

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

Let $G = \mathbb{F}_3^2$ under addition. We prove that every partition of G into parts of sizes four and five admits four pairwise disjoint cross edges whose two oriented differences exhaust $G \setminus \{0\}$. Equivalently, the four edges use the four nonzero directions of the affine plane exactly once. The proof is an exact finite computation: all $\binom{9}{4} = 126$ choices of the four-element part and all $5P_4 = 120$ injections for each choice are checked. One witness for every part is retained in a certificate, and a second implementation validates all witnesses and separately repeats the full enumeration. The result concerns only the additive group $\mathbb{F}_3^2$.

1 Statement and context

For a nonzero $d \in G$, write $[d] = \{d, -d\}$. The eight nonzero elements of G form the four directions

$$\mathcal{D} = \{[(1, 0)], [(0, 1)], [(1, 1)], [(1, 2)]\}. \quad (1)$$

If $e = \{x, y\}$ is an edge, its direction is $[x - y]$.

The cross-part question considered here is the order-nine instance of a general matching problem for odd-order finite abelian groups. A general cross-part version appears as Open Problem 2 of Affurrahman, Primaskun, Putri, and Wijaya [1]. Yip, Yoo, and Yu prove the corresponding same-part statement for arbitrary odd-order finite abelian groups and explicitly note that their method does not appear to extend directly to the cross-part version [2, Problem 1.6, Theorem 1.7, and the following paragraph]. Our comparison is limited to these sources; we make no priority claim and no assertion about a larger group.

Theorem 1. *Let $G = \mathbb{F}_3^2$, and let $G = A \dot{\cup} B$ with $\{|A|, |B|\} = \{4, 5\}$. After naming the four-element part A , there are four pairwise vertex-disjoint cross edges $\{a_i, b_i\}$, with $a_i \in A$ and $b_i \in B$, such that, as a multiset,*

$$\biguplus_{i=1}^4 \{a_i - b_i, b_i - a_i\} = G \setminus \{(0, 0)\}.$$

2 Finite reduction

Fix a four-subset $S \subset G$ and put $T = G \setminus S$. Four disjoint cross edges must use every point of S and four distinct points of T . They are therefore equivalent to an injection $f : S \rightarrow T$.

Lemma 2. *An injection $f : S \rightarrow T$ gives the matching required in Theorem 1 if and only if*

$$\{[s - f(s)] : s \in S\} = \mathcal{D}.$$

Proof. Because S and T are disjoint, $s - f(s)$ is nonzero. In characteristic three, a nonzero vector is different from its negative, so each edge $\{s, f(s)\}$ contributes precisely the two distinct oriented differences d and $-d$. Hence four edges contribute every nonzero element once if and only if their four directions are the four members of $\mathcal{D}$. Injectivity of f gives vertex-disjointness. $\square$

There are exactly $\binom{9}{4} = 126$ possible sets S . Order the nine points lexicographically and use the induced order $s_1, \dots, s_4$ on S . Every injection $S \rightarrow T$ then occurs exactly once as a length-four permutation $(t_1, \dots, t_4)$ of T , of which there are

$$5P_4 = 5 \cdot 4 \cdot 3 \cdot 2 = 120.$$

Thus the theorem is decided by the following finite predicate on exactly $126 \cdot 120 = 15120$ candidates: reduce each $s_i - t_i$ componentwise modulo three, replace it by its antipodal class, and accept precisely when the four classes are distinct. By Lemma 2, this predicate is equivalent to the desired matching condition.

3 Exact enumeration and certificate

The program `evidence/exhaustive_search.py` implements the preceding loops using only integer arithmetic, combinations, and permutations. It stores the first accepted injection for every four-subset in `evidence/exhaustive_certificate.json` and also counts every accepted injection. The complete output is summarized in Table 1.

Quantity	Exact value
Four-subsets reconstructed	126
Injections checked per four-subset	120
Total injections checked	15120
Four-subsets with 8 valid injections	54
Four-subsets with 12 valid injections	72
Minimum number of valid injections	8
Serialized witnesses	126

Table 1: Exact enumeration results.

In particular, every one of the 126 four-subsets has a certified injection. For a separate check, `evidence/independent_verify.py` imports no search code. It reconstructs the universe and all four-subsets, represents a direction by the lexicographically smaller of d and $-d$, verifies the partition, injectivity, and four-direction conditions for each serialized witness, and then independently re-enumerates all 15120 injections. Its report `evidence/independent_verification.json` records that all 126 witnesses pass and reproduces the 54/72 count distribution.

The loops cover all values of the universal quantifier, and the certificate contains an accepted witness for each value. Lemma 2 therefore converts these finite certificates into a proof of Theorem 1. No random choices, floating-point calculations, symmetry assumptions, or unproved extrapolations are used.

4 Reproduction and scope

The two programs require only the Python 3.12 standard library. From the project root, set `PYTHON` to a Python 3.12 interpreter and run

```
$PYTHON evidence/exhaustive_search.py \
  --output evidence/exhaustive_certificate.json
$PYTHON evidence/independent_verify.py \
  evidence/exhaustive_certificate.json \
  --report evidence/independent_verification.json
```

The first command regenerates the complete witness certificate; the second validates that certificate and repeats the exhaustive count by a differently written predicate.

This is a computer-assisted finite proof, not a structural classification. It proves the universal claim for $\mathbb{F}_3^2$ and the fixed part sizes $4+5$; the general cross-part problem for other odd-order groups is not addressed.

Computational and AI assistance. Exact Python programs and a machine-readable certificate were used for the finite proof. An OpenAI language model assisted with program development, checking, and preparation of the exposition and LaTeX. Model output is not used as mathematical evidence: the conclusion rests on the explicit finite reduction, exact programs, and checkable certificate described above.

References

- [1] Muhammad Afifurrahman, Devi Imulia Dian Primaskun, Pritta Etriana Putri, and Laurence P. Wijaya. On constructing 1-factors of labelled complete graph. *Boletín de la Sociedad Matemática Mexicana*, 31(2):82, 2025. doi:10.1007/s40590-025-00772-2.
- [2] Chi Hoi Yip, Semin Yoo, and Shikang Yu. Paley-type matrices and 1-factorizations of complete graphs. arXiv:2601.12250v3, 2026. Version 3, 1 September 2026.